

## Trainings ▾

### General Information

The Holset® air compressor is an engine-driven, piston-type compressor which supplies compressed air to operate air powered devices. The compressor runs continuously, but has loaded and unloaded operating modes. The operating mode is controlled by a pressure activated governor and the compressor unloading assembly. When the air system reaches cut-out pressure, the governor applies an air signal to the air compressor unloader assembly, causing the unloader cap valve to activate and stops compressed air from flowing into the air system. As the air in the air system is used, the pressure drops. A cut-in pressure, the governor exhausts the air signal to the compressor unloader assembly, allowing the compressor to again pump compressed air into the air system.

- In non-E-Type (conventional) models, the unloader cap holds the intake valve open during the unloaded mode, causing air exchange in and out of the compressor through the open intake valve.
- The E-Type (E for economy) unloader differs in that the intake is closed by the unloader cap during the unloaded mode. Back pressure on the exhaust valve at the same time traps air in the cylinder causing the piston to move against an air spring, thereby, reducing parasitic horsepower draw in the unloaded mode. Additional benefits include reduced oil passage, cooler exhaust air temperature, and unlimited turbocharge boosting capability. For more information on E-Type systems, reference the service bulletin Air Compressor Familiarization - Holset® E-Type, Bulletin 3666104.

The E-Type system can require modification to the vehicle air system plumbing if an air dryer vented to the atmosphere during the unloaded mode is used on the vehicle. All QE model air compressors utilize the E-Type unloader.

### Holset® SS191 A/C Model

The SS191 series air compressors are available in an air cooled or water cooled versions, with the difference being changes to the cylinder head to incorporate a water passage. There are no differences to the rebuild procedures for these versions.

The SS191B series air compressors are available with or without a power steering pump housing. Differences in the procedures for these versions will be shown where necessary.

